

MahjongSensei

An interactive web app that teaches Singaporean Mahjong and trains real in-game decision making.

Team **Chew An Jie** and **Dylan Wee** Stack **React**, **Django REST**, **Python engine** Prototype **Live and working**

? The Problem

Mahjong has one of the steepest learning curves of any tile game. Beginners struggle with tile recognition, hand structure, scoring, and above all deciding what to discard. Most learners also have no experienced player on hand to explain why a move was good or bad. Without that feedback, bad habits stick and progress stalls.

! Our Solution

MahjongSensei combines guided lessons, live hand analysis, and hands on practice in one place. A learner moves from reading tiles, to understanding winning hands, to getting a recommended discard, to playing a full game. A custom Python Mahjong engine powers the teaching, the analysis, and the computer opponents, so every feature speaks the same rules.

★ Core Features Built This Milestone

- 1

Interactive Tutorial
Three guided modules cover tile recognition, meld types, and winning hands. Each one ends in a quiz, and completion saves per user through a **TutorialProgress** model.
- 3

Interactive Trainer
Get dealt a hand, choose a discard, and receive instant feedback scored against the optimal play. Every attempt is logged and running accuracy is tracked.
- 5

Accounts and Profile
Registration and login use JWT tokens. Protected routes survive a refresh by restoring the session, and a profile page holds the stats foundation.

- 2

Hand Helper
Build a 14 tile hand with a visual picker and get the best tile to discard plus the reasoning. A **HandEvaluator** and **ValuationAlgorithm** test every possible discard.
- 4

Solo Play
Play a full game against three AI opponents. The engine handles dealing, draws, flowers, concealed Kongs, self drawn wins, and the washout draw.
- 6

Win Check and Tai Scoring
Detects standard, Seven Pairs, and Thirteen Orphans hands, then scores wins in tai under Singapore rules including winds, dragons, flowers, and flushes.

⚙️ How It Is Built

